

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

May 2012

EE 101 Fundamentals of Electrical & Electronics Engineering

Date: 15.05.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain the temperature co-efficient of resistance and prove the [05]

$$\alpha_{t2} = \frac{1}{\frac{1}{\alpha_{t1}} + (t_2 - t_1)}$$

- (b) Compare series circuit with parallel circuit. [02]

- Q - 2 (a) Explain voltage and current relations in delta connected system with phasor diagram. [05]

- (b) An alternating emf is represented by $e = 200 \sin 314 t$ [05]

Determine : (i) maximum value (ii) frequency (iii) time period (iv) angular frequency.

- (c) State and explain Coulomb's laws of electrostatics. [04]

OR

- Q - 2 (a) Compare similarities of Electric circuit and Magnetic circuit. [04]

- (b) Explain different methods of power measurement in three phase circuit. [06]

- (c) Two capacitors $2 \mu\text{F}$ and $4 \mu\text{F}$ are connected in series. A potential difference of 900 V is applied across the combination. Find the potential difference across each capacitor. [04]

- Q - 3 (a) State and explain (i) Fleming's right hand rule and (ii) Lenz's law. [04]

- (b) Prove that average power consumed by purely inductive is zero when a a.c. voltage is applied to it. [05]

- (c) An alternating voltage of 200V, 50 Hz is applied to a coil of negligible resistance and inductance of 0.1 H. Find the current drawn if (i) frequency is reduced to half and (ii) frequency is doubled. The supply voltage is kept constant. [05]

OR

- Q - 3 (a) Derive expressions for impedance, current and power factor for an R-C series circuit when supplied with a.c. voltage. Draw also the phasor diagram. [06]

- (b) A series circuit has resistance of 10 ohms, inductance ($200/\pi$) mH and capacitance [06]

($1000/\pi$) μF . Calculate: (i) the current flowing in the circuit of supply voltage 250 V, 50 Hz. (ii) power factor of the circuit (iii) power drawn from the supply

- (c) Define following terms: [02]
(i) Resonance (ii) Q-factor.

SECTION – II

- Q - 4 (a) Explain the operation of half wave rectifier. [05]
(b) Define following terms (i) Knee voltage (ii) Breakdown voltage [02]
Q - 5 (a) Explain zener diode working as voltage stabilizer. [07]
(b) A transistor employs a $4\text{k}\Omega$ load and $V_{cc}=13\text{V}$. What is the maximum input signal if $\beta=100$? Given $V_{knee}=1\text{V}$ and a change of 1 V in V_{BE} causes a change of 5 mA in collector current. [07]

OR

- Q - 5 (a) Explain Common base configuration in detail. [07]
(b) Explain operation of biased clipper. [07]
Q - 6 (a) Write down the advantages and disadvantages of Digital system over Analog system. [07]
(b) Explain following logic gates in detail (i) AND gate (ii) NAND gate. [07]

OR

- Q - 6 (a) Explain Zener breakdown in detail. [07]
(b) (i) Convert $(110001)_2$ to its decimal number. [01]
(ii) Convert $(24.6)_8$ to the equivalent decimal number. [02]
(iii) Convert decimal number 541 to hexadecimal. [02]
(iv) Convert $(\text{B2F})_{16}$ to octal. [02]
